Investigating the Geographical Generalizability of AI-based Wildfire Spread Prediction Models

Original Research by Jocelyn Zheng

Introduction

Recent California wildfires have sparked concern over the increasing risks of US wildfires due to human-caused activities (Westerling et al., 2006; Cunningham et al., 2024).

AI-based wildfire spread prediction models (WSPMs) are critical for emergency responses (Tymstra et al., 2020).

Geospatial AI-based models struggle to capture regional differences while generalizing across regions (Goodchild et al., 2021; Li et al., 2024).

Review of Literature

Despite spatial heterogeneity posing a problem for geospatial models, the topic has been neglected in literature about wildfire spread prediction.

Konkol, M., Kray, C., & Pfeiffer, M. (2019). Computational reproducibility in geoscientific papers: Insights from a series of studies with geoscientists and a reproduction study. International Journal of Geographical Information Science, 33(2), 408–429. https://doi.org/10.1080/13658816.2018.1508687

Next-Day Wildfire Spread is a dataset created by Google Research that aggregates wildfire data across the contiguous US.

Huot, F., Hu, R. L., Goyal, N., Sankar, T., Ihme, M., & Chen, Y.-F. (2022). Next Day Wildfire Spread: A Machine Learning Dataset to Predict Wildfire Spreading From Remote-Sensing Data. IEEE Transactions on Geoscience and Remote Sensing, 60, 1–13. https://doi.org/10.1109/TGRS.2022.3192974.

Using a U-Net-based architecture provided state-of-the-art performance in wildfire spread prediction.

Fitzgerald, J., Seefried, E., Yost, J., Pallickara, S., & Blanchard, N. (2023). Paying attention to wildfire: Using u-net with attention blocks on multimodal data for next day prediction. In Proceedings of the 25th International Conference on Multimodal Interaction. 470–480. https://doi.org/10.1145/3577190.3614116

Statement of Purpose

The purpose of the study is to evaluate how geographical imbalance affects a U-Net model's ability to generalize WSP across different regions in the contiguous US.

Hypothesis

We hypothesize that models will predict less accurately when tested on geographic regions less represented in the training data, and that models trained on geographically balanced datasets will generalize better.

Role of Mentor

- Aided with topic selection for study
- Provided guidance throughout project
- Reviewed Research Plan and Manuscript

Role of Student

- Posed research question
- Designed methodology
- Conducted literature review
- Developed hypothesis
- Trained and tested models
- Wrote manuscript
- Created presentation and poster board

Materials & Methods

- 64km x 64km samples extracted from four US regions
  - Western US
  - Southern US
  - North Central US
  - Northeast US
- All regions sampled spanned 2012-2020
- Data collected through Google Earth Engine API

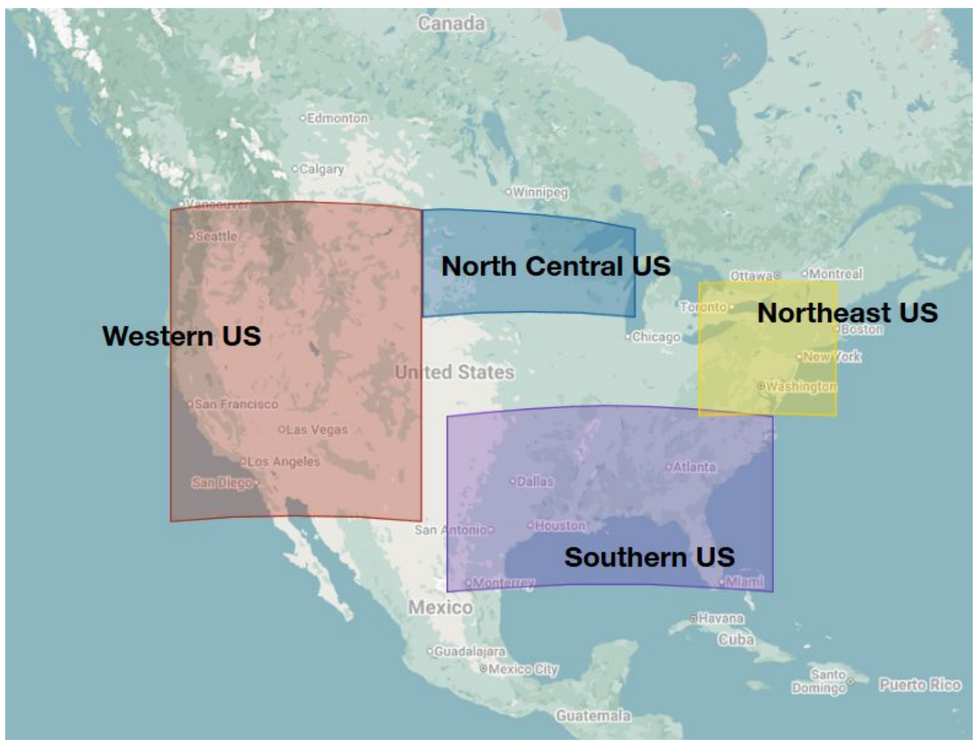

Fig. 1. A visualization of the regions where data were collected, created in Google Maps Editor.

- U-Net trained for 20 epochs; sigmoid focal loss function
  - Random 8:2 train/test split by weeks
  - Image Segmentation task

Two-part study

- Train U-Net on NDW Dataset, test on individual regions
- Create geographically balanced (GB) dataset with 3,141 samples per region, train U-Net on GB dataset, test on each region

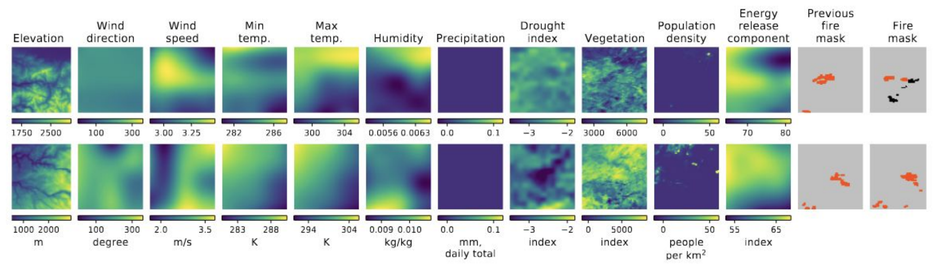

Fig. 1. Examples from our "Next Day Wildfire Spread" dataset. Each example is 64 km × 64 km at 1 km resolution. Each row corresponds to a location and time t at which a fire occurred, and the columns represent the different variables. In each row, all the weather, drought, vegetation, and population density variables are extracted at that same location at time t. "temp" denotes temperature; "wind direction" is the azimuth in degrees clockwise from north. The "previous fire mask" corresponds to the fire locations at time t, while "fire mask" corresponds to the following day at t + 1 day. In the fire masks, red corresponds to fire, and while gray corresponds to no fire. Black indicates uncertain labels (i.e., cloud coverage or other unprocessed data) (Huot et al., 2022).

Results

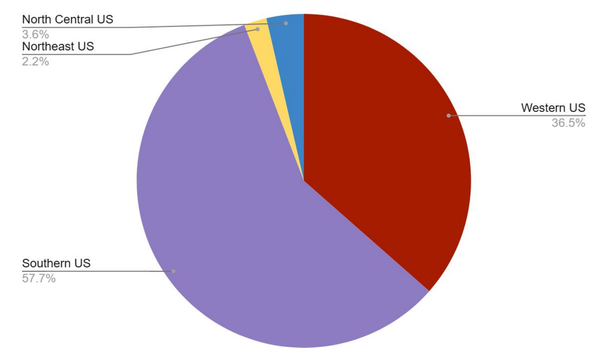

Figure 6b: Original distribution of samples per data region.

Table 6a: Distribution of Training Samples and Average Fire Pixels for each region of data

Data Region	Total Samples	Total Fire Pixels	Avg Fire Pixels per Sample
Western US	52216	2050226	39.26
Southern US	82534	886261	10.74
Northeast US	3141	30073	9.57
North Central US	5166	86441	16.73

Table 4: Performance of U-Net on selected data regions when trained on the entire Next Day Wildfire Dataset.

Tested Data Region	Precision	Recall	F1
Southern US	11.37%	3.85%	5.75%
Western US	23.89%	47.94%	31.89%
Northeastern US	0	0	0
North Central US	3.31%	0.52%	0.91%
Combined	9.64%	13.08%	9.64%

Table 5: Results of model trained on geographically balanced dataset and tested on different geographical regions.

Tested Data Region	Precision	Recall	F1
Southern US	8.90%	4.82%	6.25%
Western US	21.88%	45.84%	29.62%
Northeastern US	26.79%	7.38%	11.57%
North Central US	1.70%	0.47%	0.74%
Combined	14.82%	14.63%	12.05%

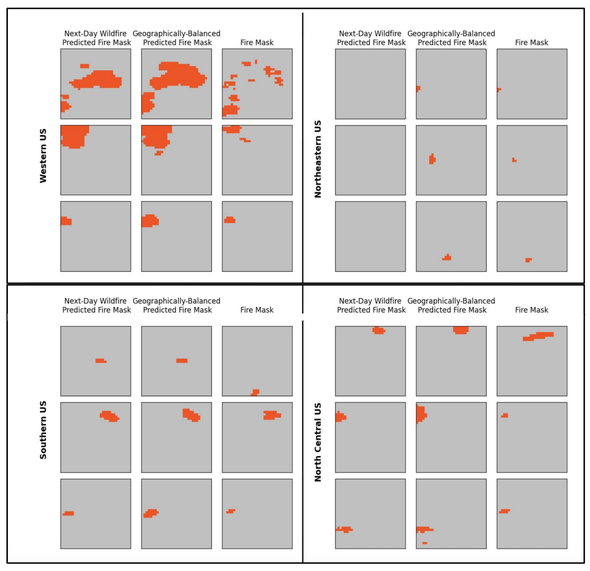

Figure 7: Examples of inferences of U-Nets trained on Next Day Wildfire Spread dataset and geographically-balanced dataset and tested on individual geographic regions.

Discussion

U-Net trained on Next-Day Wildfire Dataset achieved 0% F1 in Northeast US, but 31.89% F1 in Western US.

The geographically-balanced dataset improved the Northeast US and Southern US predictions without much decrease in the Western US and North Central US.

Geographically-balanced datasets can improve performance impacted by geographic bias.